R² Calculation, Adjusted R², Residual and Assumptions of Regression

$$Y = a + bX$$

$$Y = 10.071 + 1.0178X$$

1. Slope(b)
Represents the expected change in predicted Y for a one-unit increase in X.

Slope = 1.017857

For each additional one unit of advertising expenditure, the model predicts an average increase of approximately 1.018 in sales.

2. Intercept(a)
The intercept is the predicted value of Y when X equals zero.

our intercept is

a = 10.07144

when X = 0

$$Y = 10.071 + 1.0178*0$$

$$Y = 10.071$$

The model predicts sales of 10.07144 when advertising is zero, provided that the interpretation is meaningful within the business context.

R- Squared

It tells us what proprtion of the variation in the dependent variable is explained by the regression model.

$$R^2 = 1 - \frac{SS_{Residual}}{SS_{Total}}$$

$SS_{Residual}$ Sum of Squared differences between actual and predicted values. $\sum(Y-\hat{Y})^2$

SS_{Total} Sum of Squared differences between actual values and their mean.

Logic:

1. If predictions are close to the actual values, residual variation is small
2. If residual variation is smalll relative to total variation R^2 is higher.
3. If the model does not imporve on predicting the mean, R^2 is zero for the sual intercept based model

Important:

R^2 measures in sample fit. A high value does not prove causastion or guarantee accurate predictions on the new data.

Calcuating R-Squared

$\hat{Y} = 10.071 + 1.0178X$

Spend(X)	Sales(Y)	Predicted $\hat{Y}$	Residual$(Y-\hat{Y})$	$(Y-\hat{Y})^2$	(Y-Ybar)	$(Y-Ybar)^2$
10	20	20.249	-0.249	0.062001	-35.875	1287.016
20	28	30.427	-2.427	5.890329	-27.875	777.0156
30	42	40.605	1.395	1.946025	-13.875	192.5156
40	50	50.783	-0.783	0.613089	-5.875	34.51563
50	65	60.961	4.039	16.31352	9.125	83.26563
60	72	71.139	0.861	0.741321	16.125	260.0156
70	80	81.317	-1.317	1.734489	24.125	582.0156
80	90	91.495	-1.495	2.235025	34.125	1164.516

Residual Sum of Squares $\sum(Y-\hat{Y})^2$ 29.5358

Ybar = $\sum Y/n$ 55.875

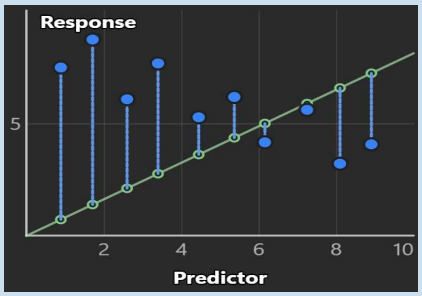
Total Sum of Squares $\sum(Y-Ybar)^2$ 4380.875

$R^2 = 1-(SSr/SSt)$

$R^2 = 1-(29.5358/4380.875)$

$R^2 = 0.993258014$

$R^2 = 0.993258033$



Example:
SSr = 120
SSt =800

$R^2 = 1-(SSr/SSt)$

0.85

The model exaplains 85% of the observed variation in Y

Adjusted R-Squared

Adjusted Rsquared evaluates model fit while accounting for the number of predictors and the sample size.

Technically:
Adjusted R-Squared adjusts ondinary R-squred using the model's residual degrees of freedom.

$$R^2_{Adjusted} = 1-(1 - R^2) * \frac{n-1}{n-k-1}$$

n = Number of observations
k = number of predictors
R^2 = ordinary R-squred

Example:

$R^2 = 0.85, n = 20, k = 2$

$$R^2_{Adjusted} = 1 - (1 - 0.85) * \frac{20-1}{20 - 2 - 1}$$

$$R^2_{Adjusted} = 0.832353$$

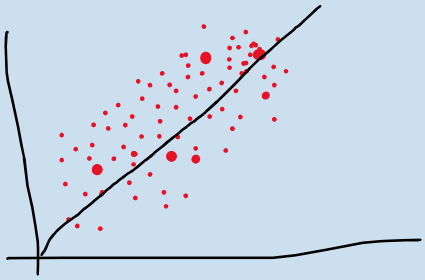
Adjusted R-Squared is approximately 83.24%

Why do we need Adjusted R-Squared?
Company predicts sales using
Model A: Advertising Spend
Model B: Advertising spend, employee shoe size and birth months
Ordinary R-squared cannot decrease when additional predcitors are added to a standard least squares model fitted to the same observations with and intercept.

Adjusteed R-squared can decrease if an additional predictor does not imporve the model sufficiently

Least Squares Regression:
A method used to find the best fitting line
Minimizing the errors(squared residuals) to position the line
A line equation Y = a+bX , Y = mx+c

Draw the line in such a way so that the distance between the points and the line is minimum.



Residuals

A residual is the difference between an actual value and the value predicted by the regression model.

$$e = Y - \hat{Y}$$

Residual:
+ve: actual value is greater than the predicted
-ve: Actual value is less than predicted
: Prediction is close to actual

Business Example:
A retailer forecasts montly sales of ₹5 lakh, but actual sales are ₹5.8 lakh

e = 5.8 - 5.0
e = +0.8 lakh
The model underpredicted sales by ₹80,000.

Possible reasons:
An unexpected promotion
Higher seasonal demad
Competitor stock shortage
Variable missing from the model.

Residual Plot:
Explain that scattered plots whose points are scattered around zero without a systematic pattern
A curved pattern may suggest normality
A funnel shaped pattern may indicate unequal error variance
Extreme points may be influential observation.

Business Application:
Residual Analysis helps identify where a sales forecasting model systematically misses actual outcomes

Assumptions of Linear regression

1. Linearity:
The mean relationship is approximately linear
2. Independence:
Errors are indpendent
3. Homoscedasticity:
Error variance us approximately constant
4. Normality of errors
Erros are approcimately normally distributed for conventional small sample inference
5. Outlier and influence:
A few unsusual observations should unduly distort the fit.

Excel Regression

Spend(X)	Sales(Y)
10	20
20	28
30	42
40	50
50	65
60	72
70	80
80	90

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.996623316
R Square	0.993258033
Adjusted I	0.992134372
Standard I	2.218697602
Observati	8

	Coefficients	*Standard Error*
Intercept	10.071429	1.728795243
Spend(X)	1.0178571	0.034235247

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	4351.339286	4351.339	883.948	9.60088E-08
Residual	6	29.53571429	4.922619		
Total	7	4380.875			

RESIDUAL OUTPUT

Observation	dicted Sales	Residuals	dard Residuals
1	20.25	-0.25	-0.12171
2	30.428571	-2.428571429	-1.1823
3	40.607143	1.392857143	0.678081
4	50.785714	-0.785714286	-0.38251
5	60.964286	4.035714286	1.964698
6	71.142857	0.857142857	0.417281
7	81.321429	-1.321428571	-0.64331
8	91.5	-1.5	-0.73024

PROBABILITY OUTPUT

Percentile	Sales(Y)
6.25	20
18.75	28
31.25	42
43.75	50
56.25	65
68.75	72
81.25	80
93.75	90

Real Life Business Applications

1. Marketing
Example: Advertising spend vs sales
Objective: Understand relationships and forecast sales

2. Retail
Example: Discount percentage vs units sold
Objective: Examine the price-promotion relationships/

3. Finance
Example: Interest rates vs borrowing demand
Objective: Analyze financial relationships.

4. HR
Example: training hours vs performance
Objective: estimate employee outcomes

5. Operations
Example: Machine operating hours vs production output
Objective: Estimate production levels

6. E - Commerce
Example: Delivery time vs customer satisfacition
Objective: Investigate service performance

7. Healthcare
Example: Treatment dosage vs measured response
Objective: Analyze relationships with appropriate study design

MINI CASE Study

An ecommerce company builds a model to estimate sales from advertising spend
The analyst discovers:
* Positive slope
* R-Squared of 0.78
* Some large positive residuals during promotional periods.

Q1. What does the positve slope indicate?
Q2. What does the R-squared of 0.78 mean?
Q3. What do the positive residuals indicate?
Q4. Why might promotional periods produce large residuals?

Expected interpretation:
The model predicts higher sales as advertising spend increases.
It explains 78% of observed sales variation in the fitted dataset.
Positive residuals mean actual sales exceeded predictions.
Promotions may be a relevant omitted variable.
The model alone does not establish that increasing advertising will cause sales to rise.